Warming Up — Solution

Setup

The solar panels face the sun perpendicularly at noon, giving maximum absorption. At other daylight hours the sun hits at an angle. The sun rises at **6 a.m.** and sets at **6 p.m.** — no energy is collected at night.

Panel area = 30 ft^2. At noon, rate = 80 W/ft^2 × 30 ft^2 = **2,400 watts**.

The Rate Function

The fraction of maximum sunlight received equals the sine of the sun's elevation angle. The sun's elevation traces a half-sine from 6 a.m. (0°) through noon (90°) to 6 p.m. (0°). So the absorption rate is:

Rate(t) = 2400 · sin(π(t – 6) / 12) watts for $6 \leq t \leq 18$

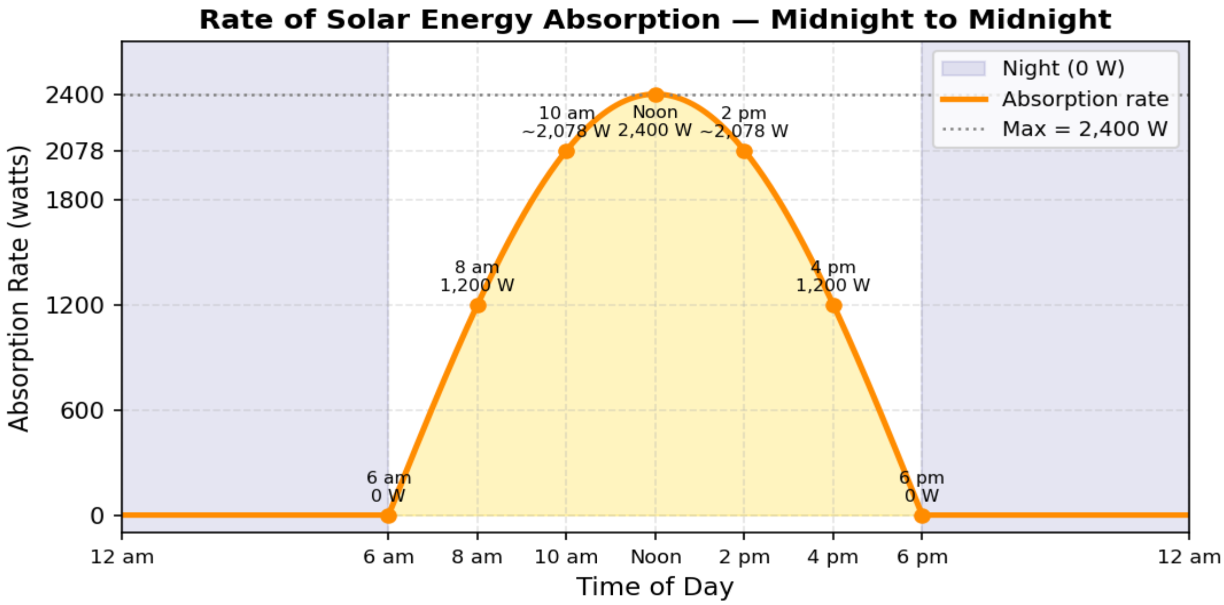
Rate(t) = 0 for $0 \leq t < 6$ and $18 < t \leq 24$

where *t* is hours after midnight.

Key Values

Time	Calculation	Rate (watts)
Midnight – 6 a.m.	No sunlight	0
6 a.m.	sin(0) = 0	0
8 a.m.	2400 · sin(π/6) = 2400 · 0.500	1,200
10 a.m.	2400 · sin(π/3) = 2400 · 0.866	≈ 2,078
Noon	2400 · sin(π/2) = 2400 · 1.000	2,400
2 p.m.	symmetric with 10 a.m.	≈ 2,078
4 p.m.	symmetric with 8 a.m.	1,200
6 p.m.	sin(π) = 0	0
6 p.m. – Midnight	No sunlight	0

Graph: Rate of Energy Absorption vs. Time of Day



What the Graph Tells Us

The graph is a **half sine-wave** from 6 a.m. to 6 p.m., flat at zero during the night. It is perfectly symmetric around noon because the sun's path is symmetric around its highest point at the equinox.

The **total energy collected** over the day equals the area under this curve — the integral of the rate from t = 6 to t = 18:

$\int_{6}^{18} 2400 \cdot \sin(\pi(t-6)/12)\,dt = 2400 \cdot (24/\pi) \approx$ **18,335 watt-hours** ≈ **18.3 kWh**